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PROVIDING DRAGGABLE SHUTTER BUTTON DURING VIDEO RECORDING

Abstract

Aspects of the present disclosure involve a system comprising a computer-readable storage medium storing a program and method for providing an draggable shutter button during video recording. The program and method provide for displaying a user interface within an application running on a device, the user interface presenting real-time image data captured by a camera of the device, the user interface including a shutter button which is configured to be selectable by a user to initiate video recording in response to a first user gesture; and upon detecting the first user gesture selecting the shutter button, initiating video recording with respect to the real-time image data, and providing for the shutter button to be draggable in predefined directions to perform respective functions related to the video recording.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application is a continuation of U.S. patent application Ser. No. 18/326,789, filed May 31, 2023, which is incorporated by reference herein in its entirety.

TECHNICAL FIELD

[0002] The present disclosure relates generally to a messaging system, including providing a draggable shutter button during video recording.

BACKGROUND

[0003] Systems such as messaging systems provide for the exchange of message content between users. For example, such systems allow users to exchange message content (e.g., text, video, pictures) between each other.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

[0004] In the drawings, which are not necessarily drawn to scale, like numerals may describe similar components in different views. To easily identify the discussion of any particular element or act, the most significant digit or digits in a reference number refer to the figure number in which that element is first introduced. Some non-limiting examples are illustrated in the figures of the accompanying drawings in which:

[0005] FIG. 1 is a diagrammatic representation of a networked environment in which the present disclosure may be deployed, according to some examples.

[0006] FIG. 2 is a diagrammatic representation of a messaging system, according to some examples, that has both client-side and server-side functionality.

[0007] FIG. 3 is a diagrammatic representation of a data structure as maintained in a database, according to some examples.

[0008] FIG. 4 is a diagrammatic representation of a message, according to some examples.

[0009] FIG. 5 is flowchart illustrating a process for providing a draggable shutter button during video recording, in accordance with some examples.

[0010] FIGS. 6A-6B illustrate an example user interface for providing an indication of video recording, in accordance with some examples.

[0011] FIG. 7 illustrates an example user interface for providing an indication of video recording together with augmented reality content, in accordance with some examples.

[0012] FIG. 8 illustrates an example user interface for providing an indication of video recording together with a preview icon, in accordance with some examples.

[0013] FIG. 9 illustrates an example user interface for providing an indication of video recording in association with a multi-clip capture mode, in accordance with some examples.

[0014] FIG. 10 illustrates example user interfaces for indicating progress of video recording, in accordance with some examples.

[0015] FIGS. 11A-11B illustrate an example user interface for providing a shutter button to be draggable in a first direction to initiate hands-free recording, in accordance with some examples.

[0016] FIGS. 12A-12B illustrate an example user interface for providing a shutter button to be draggable in a second direction to switch between a front-facing camera and a rear-facing camera,

in accordance with some examples.

[0017] FIGS. **13A-13B** illustrate an example user interface for providing a shutter button to be draggable in third and fourth directions to respectively zoom in and out, in accordance with some examples.

[0018] FIG. **14** illustrates example user interfaces for customizing an accent color used to indicate that video recording is active, in accordance with some examples.

[0019] FIG. **15** is a flowchart illustrating a process for providing a draggable shutter button during video recording, in accordance with some examples.

[0020] FIG. **16** is a diagrammatic representation of a machine in the form of a computer system within which a set of instructions may be executed to cause the machine to perform any one or more of the methodologies discussed herein, according to some examples.

[0021] FIG. **17** is a block diagram showing a software architecture within which examples may be implemented.

DETAILED DESCRIPTION

[0022] Systems such as messaging systems typically allow users to exchange media content items (e.g., messages, images and/or video) with one another. As described herein, the system provides for a user to capture video, and provides for facilitated access to functions related to video recording while video is being recorded.

[0023] The disclosed embodiments relate to a system which provides a shutter button to be draggable during video recording. The shutter button is draggable in different directions to perform respective functions related to video recording. In response to a press-and-hold gesture of the shutter button, the interaction client **104** initiates video recording. The interaction client **104** also presents a hands-free icon (e.g., a lock icon) for hands-free video recording, and a flip camera icon for switching between front and rear-facing cameras while recording. The interaction client **104** provides for the shutter button to be draggable in: a first direction (e.g., left, towards the hands-free icon) to initiate hands-free recording; a second direction (e.g., right, towards the flip camera icon) to switch between the front-facing camera and the rear-facing camera; a third direction (e.g., upward) to zoom in on image data captured in real-time; and a fourth direction (e.g., downward) to zoom out with respect to the captured image data.

[0024] By virtue of providing a shutter button that is draggable in different directions for different functions, the system provides increased interactive feedback and facilitates video recording. Without providing such a draggable shutter button, performing the various functions may be difficult and/or cumbersome during video recording. Thus, the interaction client **104** facilitates the capturing of videos, thereby saving time for the user (e.g., by reducing the number of retakes), and reducing computational resources/processing power for the interaction system **100**.

Networked Computing Environment

[0025] FIG. **1** is a block diagram showing an example interaction system **100** for facilitating interactions (e.g., exchanging text messages, conducting text audio and video calls, or playing games) over a network. The interaction system **100** includes multiple client systems **102**, each of which hosts multiple applications, including an interaction client **104** and other applications **106**. Each interaction client **104** is communicatively coupled, via one or more communication networks including a network **108** (e.g., the Internet), to other instances of the interaction client **104** (e.g., hosted on respective other user systems **102**), an interaction server system **110** and third-party servers **112**). An interaction client **104** can also communicate with locally hosted applications **106** using Applications Program Interfaces (APIs).

[0026] Each user system **102** may include multiple user devices, such as a mobile device **114**, head-wearable apparatus **116**, and a computer client device **118** that are communicatively connected to exchange data and messages.

[0027] An interaction client **104** interacts with other interaction clients **104** and with the interaction server system **110** via the network **108**. The data exchanged between the interaction clients **104**

(e.g., interactions **120**) and between the interaction clients **104** and the interaction server system **110** includes functions (e.g., commands to invoke functions) and payload data (e.g., text, audio, video, or other multimedia data).

[0028] The interaction server system **110** provides server-side functionality via the network **108** to the interaction clients **104**. While certain functions of the interaction system **100** are described herein as being performed by either an interaction client **104** or by the interaction server system **110**, the location of certain functionality either within the interaction client **104** or the interaction server system **110** may be a design choice. For example, it may be technically preferable to initially deploy particular technology and functionality within the interaction server system **110** but to later migrate this technology and functionality to the interaction client **104** where a user system **102** has sufficient processing capacity.

[0029] The interaction server system **110** supports various services and operations that are provided to the interaction clients **104**. Such operations include transmitting data to, receiving data from, and processing data generated by the interaction clients **104**. This data may include message content, client device information, geolocation information, media augmentation and overlays, message content persistence conditions, social network information, and live event information. Data exchanges within the interaction system **100** are invoked and controlled through functions available via user interfaces (UIs) of the interaction clients **104**.

[0030] Turning now specifically to the interaction server system **110**, an Application Program Interface (API) server **122** is coupled to and provides programmatic interfaces to interaction servers **124**, making the functions of the interaction servers **124** accessible to interaction clients **104**, other applications **106** and third-party server **112**. The interaction servers **124** are communicatively coupled to a database server **126**, facilitating access to a database **128** that stores data associated with interactions processed by the interaction servers **124**. Similarly, a web server **130** is coupled to the interaction servers **124** and provides web-based interfaces to the interaction servers **124**. To this end, the web server **130** processes incoming network requests over the Hypertext Transfer Protocol (HTTP) and several other related protocols.

[0031] The Application Program Interface (API) server **122** receives and transmits interaction data (e.g., commands and message payloads) between the interaction servers **124** and the client systems **102** (and, for example, interaction clients **104** and other application **106**) and the third-party server **112**. Specifically, the Application Program Interface (API) server **122** provides a set of interfaces (e.g., routines and protocols) that can be called or queried by the interaction client **104** and other applications **106** to invoke functionality of the interaction servers **124**. The Application Program Interface (API) server **122** exposes various functions supported by the interaction servers **124**, including account registration; login functionality; the sending of interaction data, via the interaction servers **124**, from a particular interaction client **104** to another interaction client **104**; the communication of media files (e.g., images or video) from an interaction client **104** to the interaction servers **124**; the settings of a collection of media data (e.g., a story); the retrieval of a list of friends of a user of a user system **102**; the retrieval of messages and content; the addition and deletion of entities (e.g., friends) to an entity graph (e.g., a social graph); the location of friends within a social graph; and opening an application event (e.g., relating to the interaction client **104**).

[0032] The interaction servers **124** host multiple systems and subsystems, described below with reference to FIG. 2.

System Architecture

[0033] FIG. 2 is a block diagram illustrating further details regarding the interaction system **100**, according to some examples. Specifically, the interaction system **100** is shown to comprise the interaction client **104** and the interaction servers **124**. The interaction system **100** embodies multiple subsystems, which are supported on the client-side by the interaction client **104** and on the server-side by the interaction servers **124**. Example subsystems are discussed below.

[0034] An image processing system **202** provides various functions that enable a user to capture

and augment (e.g., annotate or otherwise modify or edit) media content associated with a message. [0035] A camera system **204** includes control software (e.g., in a camera application) that interacts with and controls hardware camera hardware (e.g., directly or via operating system controls) of the user system **102** to modify and augment real-time images captured and displayed via the interaction client **104**.

[0036] The augmentation system **206** provides functions related to the generation and publishing of augmentations (e.g., media overlays) for images captured in real-time by cameras of the user system **102** or retrieved from memory of the user system **102**. For example, the augmentation system **206** operatively selects, presents, and displays media overlays (e.g., an image filter or an image lens) to the interaction client **104** for the augmentation of real-time images received via the camera system **204** or stored images retrieved from memory of the user system **102**. These augmentations are selected by the augmentation system **206** and presented to a user of an interaction client **104**, based on a number of inputs and data, such as for example: [0037] Geolocation of the user system **102**; and [0038] Social network information of the user of the user system **102**.

[0039] An augmentation may include audio and visual content and visual effects. Examples of audio and visual content include pictures, texts, logos, animations, and sound effects. An example of a visual effect includes color overlaying. The audio and visual content or the visual effects can be applied to a media content item (e.g., a photo or video) at user system **102** for communication in a message, or applied to video content, such as a video content stream or feed transmitted from an interaction client **104**. As such, the image processing system **202** may interact with, and support, the various subsystems of the communication system **208**, such as the messaging system **210** and the video communication system **212**.

[0040] A media overlay may include text or image data that can be overlaid on top of a photograph taken by the user system **102** or a video stream produced by the user system **102**. In some examples, the media overlay may be a location overlay (e.g., Venice beach), a name of a live event, or a name of a merchant overlay (e.g., Beach Coffee House). In further examples, the image processing system **202** uses the geolocation of the user system **102** to identify a media overlay that includes the name of a merchant at the geolocation of the user system **102**. The media overlay may include other indicia associated with the merchant. The media overlays may be stored in the databases **128** and accessed through the database server **126**.

[0041] The image processing system **202** provides a user-based publication platform that enables users to select a geolocation on a map and upload content associated with the selected geolocation. The user may also specify circumstances under which a particular media overlay should be offered to other users. The image processing system **202** generates a media overlay that includes the uploaded content and associates the uploaded content with the selected geolocation.

[0042] The augmentation creation system **214** supports augmented reality developer platforms and includes an application for content creators (e.g., artists and developers) to create and publish augmentations (e.g., augmented reality experiences) of the interaction client **104**. The augmentation creation system **214** provides a library of built-in features and tools to content creators including, for example custom shaders, tracking technology, and templates.

[0043] In some examples, the augmentation creation system **214** provides a merchant-based publication platform that enables merchants to select a particular augmentation associated with a geolocation via a bidding process. For example, the augmentation creation system **214** associates a media overlay of the highest bidding merchant with a corresponding geolocation for a predefined amount of time.

[0044] A communication system **208** is responsible for enabling and processing multiple forms of communication and interaction within the interaction system **100** and includes a messaging system **210**, an audio communication system **216**, and a video communication system **212**. The messaging system **210** is responsible for enforcing the temporary or time-limited access to content by the

interaction clients **104**. The audio communication system **216** enables and supports audio communications (e.g., real-time audio chat) between multiple interaction clients **104**. Similarly, the video communication system **212** enables and supports video communications (e.g., real-time video chat) between multiple interaction clients **104**.

[0045] A user management system **218** is operationally responsible for the management of user data and profiles, and includes a social network system **220** that maintains information regarding relationships between users of the interaction system **100**.

[0046] A collection management system **222** is operationally responsible for managing sets or collections of media (e.g., collections of text, image video, and audio data). A collection of content (e.g., messages, including images, video, text, and audio) may be organized into an “event gallery” or an “event story.” Such a collection may be made available for a specified time period, such as the duration of an event to which the content relates. For example, content relating to a music concert may be made available as a “story” for the duration of that music concert. The collection management system **222** may also be responsible for publishing an icon that provides notification of a particular collection to the user interface of the interaction client **104**. The collection management system **222** includes a curation function that allows a collection manager to manage and curate a particular collection of content. For example, the curation interface enables an event organizer to curate a collection of content relating to a specific event (e.g., delete inappropriate content or redundant messages). Additionally, the collection management system **222** employs machine vision (or image recognition technology) and content rules to curate a content collection automatically. In certain examples, compensation may be paid to a user to include user-generated content into a collection. In such cases, the collection management system **222** operates to automatically make payments to such users to use their content.

[0047] A map system **224** provides various geographic location functions and supports the presentation of map-based media content and messages by the interaction client **104**. For example, the map system **224** enables the display of user icons or avatars (e.g., stored in profile data **302**) on a map to indicate a current or past location of “friends” of a user, as well as media content (e.g., collections of messages including photographs and videos) generated by such friends, within the context of a map. For example, a message posted by a user to the interaction system **100** from a specific geographic location may be displayed within the context of a map at that particular location to “friends” of a specific user on a map interface of the interaction client **104**. A user can furthermore share his or her location and status information (e.g., using an appropriate status avatar) with other users of the interaction system **100** via the interaction client **104**, with this location and status information being similarly displayed within the context of a map interface of the interaction client **104** to selected users.

[0048] A game system **226** provides various gaming functions within the context of the interaction client **104**. The interaction client **104** provides a game interface providing a list of available games that can be launched by a user within the context of the interaction client **104** and played with other users of the interaction system **100**. The interaction system **100** further enables a particular user to invite other users to participate in the play of a specific game by issuing invitations to such other users from the interaction client **104**. The interaction client **104** also supports audio, video, and text messaging (e.g., chats) within the context of gameplay, provides a leaderboard for the games, and also supports the provision of in-game rewards (e.g., coins and items).

[0049] An external resource system **228** provides an interface for the interaction client **104** to communicate with remote servers (e.g., third-party servers **112**) to launch or access external resources, i.e., applications or applets. Each third-party server **112** hosts, for example, a markup language (e.g., HTML5) based application or a small-scale version of an application (e.g., game, utility, payment, or ride-sharing application). The interaction client **104** may launch a web-based resource (e.g., application) by accessing the HTML5 file from the third-party servers **112** associated with the web-based resource. Applications hosted by third-party servers **112** are

programmed in JavaScript leveraging a Software Development Kit (SDK) provided by the interaction servers **124**. The SDK includes Application Programming Interfaces (APIs) with functions that can be called or invoked by the web-based application. The interaction servers **124** host a JavaScript library that provides a given external resource access to specific user data of the interaction client **104**. HTML5 is an example of technology for programming games, but applications and resources programmed based on other technologies can be used.

[0050] To integrate the functions of the SDK into the web-based resource, the SDK is downloaded by the third-party server **112** from the interaction servers **124** or is otherwise received by the third-party server **112**. Once downloaded or received, the SDK is included as part of the application code of a web-based external resource. The code of the web-based resource can then call or invoke certain functions of the SDK to integrate features of the interaction client **104** into the web-based resource.

[0051] The SDK stored on the interaction server system **110** effectively provides the bridge between an external resource (e.g., applications **106** or applets) and the interaction client **104**. This gives the user a seamless experience of communicating with other users on the interaction client **104** while also preserving the look and feel of the interaction client **104**. To bridge communications between an external resource and an interaction client **104**, the SDK facilitates communication between third-party servers **112** and the interaction client **104**. A Web ViewJavaScriptBridge running on a user system **102** establishes two one-way communication channels between an external resource and the interaction client **104**. Messages are sent between the external resource and the interaction client **104** via these communication channels asynchronously. Each SDK function invocation is sent as a message and callback. Each SDK function is implemented by constructing a unique callback identifier and sending a message with that callback identifier.

[0052] By using the SDK, not all information from the interaction client **104** is shared with third-party servers **112**. The SDK limits which information is shared based on the needs of the external resource. Each third-party server **112** provides an HTML5 file corresponding to the web-based external resource to interaction servers **124**. The interaction servers **124** can add a visual representation (such as a box art or other graphic) of the web-based external resource in the interaction client **104**. Once the user selects the visual representation or instructs the interaction client **104** through a GUI of the interaction client **104** to access features of the web-based external resource, the interaction client **104** obtains the HTML5 file and instantiates the resources to access the features of the web-based external resource.

[0053] The interaction client **104** presents a graphical user interface (e.g., a landing page or title screen) for an external resource. During, before, or after presenting the landing page or title screen, the interaction client **104** determines whether the launched external resource has been previously authorized to access user data of the interaction client **104**. In response to determining that the launched external resource has been previously authorized to access user data of the interaction client **104**, the interaction client **104** presents another graphical user interface of the external resource that includes functions and features of the external resource. In response to determining that the launched external resource has not been previously authorized to access user data of the interaction client **104**, after a threshold period of time (e.g., 3 seconds) of displaying the landing page or title screen of the external resource, the interaction client **104** slides up (e.g., animates a menu as surfacing from a bottom of the screen to a middle or other portion of the screen) a menu for authorizing the external resource to access the user data. The menu identifies the type of user data that the external resource will be authorized to use. In response to receiving a user selection of an accept option, the interaction client **104** adds the external resource to a list of authorized external resources and allows the external resource to access user data from the interaction client **104**. The external resource is authorized by the interaction client **104** to access the user data under an OAuth 2 framework.

[0054] The interaction client **104** controls the type of user data that is shared with external

resources based on the type of external resource being authorized. For example, external resources that include full-scale applications (e.g., an application **106**) are provided with access to a first type of user data (e.g., two-dimensional avatars of users with or without different avatar characteristics). As another example, external resources that include small-scale versions of applications (e.g., web-based versions of applications) are provided with access to a second type of user data (e.g., payment information, two-dimensional avatars of users, three-dimensional avatars of users, and avatars with various avatar characteristics). Avatar characteristics include different ways to customize a look and feel of an avatar, such as different poses, facial features, clothing, and so forth.

[0055] An advertisement system **230** operationally enables the purchasing of advertisements by third parties for presentation to end-users via the interaction clients **104** and also handles the delivery and presentation of these advertisements.

Data Architecture

[0056] FIG. **3** is a schematic diagram illustrating data structures **300**, which may be stored in the database **304** of the interaction server system **110**, according to certain examples. While the content of the database **304** is shown to comprise multiple tables, it will be appreciated that the data could be stored in other types of data structures (e.g., as an object-oriented database).

[0057] The database **304** includes message data stored within a message table **306**. This message data includes, for any particular message, at least message sender data, message recipient (or receiver) data, and a payload. Further details regarding information that may be included in a message, and included within the message data stored in the message table **306**, are described below with reference to FIG. **3**.

[0058] An entity table **308** stores entity data, and is linked (e.g., referentially) to an entity graph **310** and profile data **302**. Entities for which records are maintained within the entity table **308** may include individuals, corporate entities, organizations, objects, places, events, and so forth. Regardless of entity type, any entity regarding which the interaction server system **110** stores data may be a recognized entity. Each entity is provided with a unique identifier, as well as an entity type identifier (not shown).

[0059] The entity graph **310** stores information regarding relationships and associations between entities. Such relationships may be social, professional (e.g., work at a common corporation or organization), interest-based, or activity-based, merely for example. Certain relationships between entities may be unidirectional, such as a subscription by an individual user to digital content of a commercial or publishing user (e.g., a newspaper or other digital media outlet, or a brand). Other relationships may be bidirectional, such as a “friend” relationship between individual users of the interaction system **100**.

[0060] Certain permissions and relationships may be attached to each relationship, and also to each direction of a relationship. For example, a bidirectional relationship (e.g., a friend relationship between individual users) may include authorization for the publication of digital content items between the individual users, but may impose certain restrictions or filters on the publication of such digital content items (e.g., based on content characteristics, location data or time of day data). Similarly, a subscription relationship between an individual user and a commercial user may impose different degrees of restrictions on the publication of digital content from the commercial user to the individual user, and may significantly restrict or block the publication of digital content from the individual user to the commercial user. A particular user, as an example of an entity, may record certain restrictions (e.g., by way of privacy settings) in a record for that entity within the entity table **308**. Such privacy settings may be applied to all types of relationships within the context of the interaction system **100**, or may selectively be applied to certain types of relationships.

[0061] The profile data **302** stores multiple types of profile data about a particular entity. The profile data **302** may be selectively used and presented to other users of the interaction system **100**

based on privacy settings specified by a particular entity. Where the entity is an individual, the profile data **302** includes, for example, a user name, telephone number, address, settings (e.g., notification and privacy settings), as well as a user-selected avatar representation (or collection of such avatar representations). A particular user may then selectively include one or more of these avatar representations within the content of messages communicated via the interaction system **100**, and on map interfaces displayed by interaction clients **104** to other users. The collection of avatar representations may include “status avatars,” which present a graphical representation of a status or activity that the user may select to communicate at a particular time.

[0062] Where the entity is a group, the profile data **302** for the group may similarly include one or more avatar representations associated with the group, in addition to the group name, members, and various settings (e.g., notifications) for the relevant group.

[0063] The database **304** also stores augmentation data, such as overlays or filters, in an augmentation table **312**. The augmentation data is associated with and applied to videos (for which data is stored in a video table **314**) and images (for which data is stored in an image table **316**).

[0064] Filters, in some examples, are overlays that are displayed as overlaid on an image or video during presentation to a recipient user. Filters may be of various types, including user-selected filters from a set of filters presented to a sending user by the interaction client **104** when the sending user is composing a message. Other types of filters include geolocation filters (also known as geo-filters), which may be presented to a sending user based on geographic location. For example, geolocation filters specific to a neighborhood or special location may be presented within a user interface by the interaction client **104**, based on geolocation information determined by a Global Positioning System (GPS) unit of the user system **102**.

[0065] Another type of filter is a data filter, which may be selectively presented to a sending user by the interaction client **104** based on other inputs or information gathered by the user system **102** during the message creation process. Examples of data filters include current temperature at a specific location, a current speed at which a sending user is traveling, battery life for a user system **102**, or the current time.

[0066] Other augmentation data that may be stored within the image table **316** includes augmented reality content items (e.g., corresponding to applying “lenses” or augmented reality experiences). An augmented reality content item may be a real-time special effect and sound that may be added to an image or a video.

[0067] A story table **318** stores data regarding collections of messages and associated image, video, or audio data, which are compiled into a collection (e.g., a story or a gallery). The creation of a particular collection may be initiated by a particular user (e.g., each user for which a record is maintained in the entity table **308**). A user may create a “personal story” in the form of a collection of content that has been created and sent/broadcast by that user. To this end, the user interface of the interaction client **104** may include an icon that is user-selectable to enable a sending user to add specific content to his or her personal story.

[0068] A collection may also constitute a “live story,” which is a collection of content from multiple users that is created manually, automatically, or using a combination of manual and automatic techniques. For example, a “live story” may constitute a curated stream of user-submitted content from various locations and events. Users whose client devices have location services enabled and are at a common location event at a particular time may, for example, be presented with an option, via a user interface of the interaction client **104**, to contribute content to a particular live story. The live story may be identified to the user by the interaction client **104**, based on his or her location. The end result is a “live story” told from a community perspective.

[0069] A further type of content collection is known as a “location story,” which enables a user whose user system **102** is located within a specific geographic location (e.g., on a college or university campus) to contribute to a particular collection. In some examples, a contribution to a location story may employ a second degree of authentication to verify that the end-user belongs to

a specific organization or other entity (e.g., is a student on the university campus).

[0070] As mentioned above, the video table **314** stores video data that, in some examples, is associated with messages for which records are maintained within the message table **306**. Similarly, the image table **316** stores image data associated with messages for which message data is stored in the entity table **308**. The entity table **308** may associate various augmentations from the augmentation table **312** with various images and videos stored in the image table **316** and the video table **314**.

Data Communications Architecture

[0071] FIG. **4** is a schematic diagram illustrating a structure of a message **400**, according to some examples, generated by an interaction client **104** for communication to a further interaction client **104** via the interaction servers **124**. The content of a particular message **400** is used to populate the message table **306** stored within the database **304**, accessible by the interaction servers **124**. Similarly, the content of a message **400** is stored in memory as “in-transit” or “in-flight” data of the user system **102** or the interaction servers **124**. A message **400** is shown to include the following example components: [0072] Message identifier **402**: a unique identifier that identifies the message **400**. [0073] Message text payload **404**: text, to be generated by a user via a user interface of the user system **102**, and that is included in the message **400**. [0074] Message image payload **406**: image data, captured by a camera component of a user system **102** or retrieved from a memory component of a user system **102**, and that is included in the message **400**. Image data for a sent or received message **400** may be stored in the image table **316**. [0075] Message video payload **408**: video data, captured by a camera component or retrieved from a memory component of the user system **102**, and that is included in the message **400**. Video data for a sent or received message **400** may be stored in the image table **316**. [0076] Message audio payload **410**: audio data, captured by a microphone or retrieved from a memory component of the user system **102**, and that is included in the message **400**. [0077] Message augmentation data **412**: augmentation data (e.g., filters, stickers, or other annotations or enhancements) that represents augmentations to be applied to message image payload **406**, message video payload **408**, or message audio payload **410** of the message **400**. Augmentation data for a sent or received message **400** may be stored in the augmentation table **312**. [0078] Message duration parameter **414**: parameter value indicating, in seconds, the amount of time for which content of the message (e.g., the message image payload **406**, message video payload **408**, message audio payload **410**) is to be presented or made accessible to a user via the interaction client **104**. [0079] Message geolocation parameter **416**: geolocation data (e.g., latitudinal and longitudinal coordinates) associated with the content payload of the message. Multiple message geolocation parameter **416** values may be included in the payload, each of these parameter values being associated with respect to content items included in the content (e.g., a specific image within the message image payload **406**, or a specific video in the message video payload **408**). [0080] Message story identifier **418**: identifier values identifying one or more content collections (e.g., “stories” identified in the story table **318**) with which a particular content item in the message image payload **406** of the message **400** is associated. For example, multiple images within the message image payload **406** may each be associated with multiple content collections using identifier values. [0081] Message tag **420**: each message **400** may be tagged with multiple tags, each of which is indicative of the subject matter of content included in the message payload. For example, where a particular image included in the message image payload **406** depicts an animal (e.g., a lion), a tag value may be included within the message tag **420** that is indicative of the relevant animal. Tag values may be generated manually, based on user input, or may be automatically generated using, for example, image recognition. [0082] Message sender identifier **422**: an identifier (e.g., a messaging system identifier, email address, or device identifier) indicative of a user of the user system **102** on which the message **400** was generated and from which the message **400** was sent. [0083] Message receiver identifier **424**: an identifier (e.g., a messaging system identifier, email address, or device identifier) indicative of a user of the user system **102** to

which the message **400** is addressed.

[0084] The contents (e.g., values) of the various components of message **400** may be pointers to locations in tables within which content data values are stored. For example, an image value in the message image payload **406** may be a pointer to (or address of) a location within an image table **316**. Similarly, values within the message video payload **408** may point to data stored within an image table **316**, values stored within the message augmentation data **412** may point to data stored in an augmentation table **312**, values stored within the message story identifier **418** may point to data stored in a story table **318**, and values stored within the message sender identifier **422** and the message receiver identifier **424** may point to user records stored within an entity table **308**.

[0085] FIG. **5** is flowchart illustrating a process **500** for providing a draggable shutter button during video recording, in accordance with some examples. For explanatory purposes, the process **500** is primarily described herein with reference to the interaction client **104** of FIG. **1**. However, one or more blocks (or operations) of the process **500** may be performed by one or more other components, and/or by other suitable devices. Further for explanatory purposes, the blocks (or operations) of the process **500** are described herein as occurring in serial, or linearly. However, multiple blocks (or operations) of the process **500** may occur in parallel or concurrently. In addition, the blocks (or operations) of the process **500** need not be performed in the order shown and/or one or more blocks (or operations) of the process **500** need not be performed and/or can be replaced by other operations. The process **500** may be terminated when its operations are completed. In addition, the process **500** may correspond to a method, a procedure, an algorithm, etc.

[0086] In example embodiments, the interaction client **104** is running on a user system **102** associated with a user of the interaction system **100**. The user is associated with a user account of the interaction system **100** in conjunction with the user management system **218**. For example, the user is identified by the user management system **218** based on one or more unique identifiers (e.g., a messaging system identifier, email address and/or a device identifier) associated with the user accounts for the user. In some embodiments, the user management system **218** implements and/or works in conjunction with the social network system **220** which is configured to identify other users (e.g., friends) with which the user of the interaction client **104** has relationships.

[0087] As described herein, the interaction client **104** provides for a shutter button to be draggable in different directions to perform respective functions associated with video recording. In response to a press-and-hold gesture of the shutter button, the interaction client **104** initiates video recording. The interaction client **104** also presents a hands-free icon (e.g., a lock icon) for hands-free video recording, and a flip camera icon for switching between front and rear-facing cameras while recording. The interaction client **104** provides for the shutter button to be draggable in: a first direction (e.g., left, towards the hands-free icon) to initiate hands-free recording; a second direction (e.g., right, towards the flip camera icon) to switch between the front-facing camera and the rear-facing camera; a third direction (e.g., upward) to zoom in on image data captured in real-time; and a fourth direction (e.g., downward) to zoom out with respect to the captured image data.

[0088] Thus, following start block **502**, the interaction client **104** displays a user interface with image data and a shutter button (block **504**). In example embodiments, the interaction client **104** activates a camera (e.g., front-facing camera, or rear-facing camera) of the user system **102** upon startup. The interaction client **104** provides for displaying, within the user interface, the real-time image data captured by the camera.

[0089] In example embodiments, the interaction client **104** also displays a first set of interface elements in conjunction with the real-time image data. By way of non-limiting example, the first set of interface elements includes one or more of: a profile button for surfacing a user profile, a search button for performing various types of searches with respect to the interaction system **100**, a memories button for accessing saved media content items, a carousel interface launch button for surfacing a carousel interface for augmented reality content, a toolbar for activating a flash for the

camera or for switching between front and rear facing cameras, or a tab bar for navigating between tabs of the interaction client **104**.

[0090] In a case where the user selects the carousel interface launch button, the interaction client **104** updates the user interface to include a carousel interface. The carousel interface allows the user to cycle through and/or select different augmented reality content items, represented by respective icons, to apply with respect to the captured image data. Each augmented reality content item provides for adding a real-time special effect and/or sound to the captured image data.

[0091] In a case where no augmented reality content item is selected, the shutter button appears in a default state. For example, the shutter button appears as a circle with a border (e.g., white border) and no fill. Otherwise, the shutter button includes the respective icon, representing the currently-selected augmented reality content item, centered within the circle.

[0092] The shutter button is user-selectable to take a picture (e.g., in response to a press/tap gesture) or to record a video (e.g., in response to a press-and-hold gesture) of the image data captured by the camera. In case an augmented reality content item is selected, the picture or video is modified to include corresponding augmented reality content (e.g., a real-time special effect and/or sound). After completion of the picture or recording the video, the interaction client **104** directs to a message preview interface for generating a media content item (e.g., corresponding to the picture or video with augmented reality content, if applicable), for example, to send to friends, include in a Story, and the like.

[0093] In the example of FIG. 5, the interaction client **104** detects user selection of the shutter button (block **506**). At decision block **508**, the interaction client **104** determines whether the user selection from block **506** is a press-and-hold gesture (e.g., for video recording) or a tap gesture (e.g., for taking a picture). In example embodiments, the interaction client **104** makes such a determination via appropriate API calls with an operating system of the user system **102**. For example, a press-and-hold gesture is detectable by the operating system based on a user touching and holding the shutter button for at least a threshold period of time (e.g., ≥ 0.5 seconds). On the other hand, a tap gesture is detectable by the operating system based on the user touching shutter button for less than the threshold period of time (e.g., < 0.5 seconds).

[0094] If the user selection at decision block **508** is a press-and-hold gesture, the interaction client **104** causes the user system **102** to initiate video recording (block **514**). As noted above, the video recording is based on the real-time image data that is captured and displayed on the user system **102** (e.g., together with augmented reality content, if applicable).

[0095] At block **516**, the interaction client **104** provides a multiple indications that video recording is active. For example, the interaction client **104** replaces the first set of interface elements with a second set of interface elements within the user interface. As noted above, the first set of interface elements includes one or more of a profile button, a search button, a memories button, a carousel interface launch button, a toolbar or a tab bar. In example embodiments, the interaction client **104** causes the first set of interface elements to fade out in association with being replaced by the second set of interface elements.

[0096] By way of non-limiting example, the second set of interface elements includes one or more of: a border for the camera viewport (e.g., a border for the frame/portion of the device screen displaying the real-time image data); an animated icon within the user interface; a hands-free icon which is user-selectable to initiate hands-free recording; and/or a flip camera icon which is user-selectable icon for switching between a front-facing camera and a rear-facing camera while the video recording is active.

[0097] With respect to the border, the interaction client **104** presents the border as a solid color (e.g., yellow) along the periphery of the camera viewport. For example, the border is scaled up (e.g., grows from first size to a second size) within the user interface. In addition, the interaction client **104** displays a ring flash effect (e.g., glowing halo) along the periphery of the camera viewport and displays the border (e.g., solid yellow) on top of the ring flash effect.

[0098] In example embodiments, the interaction client **104** presents a “bounce” animation with respect to the camera viewport. For example, the bounce animation scales down the camera viewport from its original size to a smaller size, and then scales up the camera viewport from the smaller size to its original size. The border is displayed upon completion of the bounce animation. [0099] Regarding the animated icon, the interaction client **104** scales up a predefined icon within a predefined area of the user interface (e.g., top center and below the border). The interaction client **104** causes the predefined icon to blink within a predefined area of the user interface. For example, the predefined icon scales up to 30×30 pixels, and blinks every second (e.g., where blinking includes changing opacity from 0% to 100% in 0.4 s, pausing for 3 s, and changing opacity from 100% to 0% in 0.3 s). For example, the predefined icon is a ghost icon, a smiley face icon, a logo such as a company logo, or another type of icon.

[0100] With respect to the icon for hands-free recording icon, the interaction client **104** scales up this icon within a predefined area of the user interface (e.g., to the left of the shutter button). Moreover, the interaction client **104** scales up the flip camera icon within a predefined area of the user interface (e.g., to the right of the shutter button).

[0101] Moreover, if the user selection at decision block **508** is a press-and-hold gesture, the interaction client **104** updates the shutter button within the user interface. As noted above, in a case where no augmented reality content item is selected, the shutter button initially appears as a circle with a border (e.g., white border) and no fill. In example embodiments, updating the appearance of the shutter button includes scaling up the circle and changing the entire circle into a solid color (e.g., yellow).

[0102] As noted above, in a case where an augmented reality content item is selected, the shutter button initially includes the respective icon centered within the circle. Updating the appearance of the shutter button includes scaling up the circle, changing the entire circle into the solid color (e.g., yellow), and maintaining the icon centered within the circle on top of the solid color.

[0103] In example embodiments, updating the appearance of the shutter button further includes presenting a progress indicator along a periphery of the shutter button. The progress indicator provides a visual indication that the video recording is active and indicates progress of the video recording. For example, the progress indicator is displayed as an animation in which a carved-out segment positioned at the edge of the circle, travels around the circle in a clockwise manner and increases in size. The interaction client **104** provides for each full rotation of the carved-out segment around the circle to take a preset amount of time (e.g., 10 seconds to travel around the circle), with the carved-out segment resetting after each rotation.

[0104] As discussed further below with respect to FIG. **8**, upon completion of each full rotation, the interaction client **104** displays a preview icon associated with that video segment having the preset time length (e.g., 10 seconds). In example embodiments, the preview icon is a single video frame within the video segment that is automatically selected by the interaction client **104** and/or the interaction server system **110**. Additional video segments of the preset time length (e.g., 10 seconds) are added as preview icons within the user interface, to create a series of video segments. For example, the interaction client **104** provides a visual approximation of the length of the video during recording (e.g., approximated as the number of preview icons×the preset time length).

[0105] As discussed further below with respect to FIG. **9**, the interaction client **104** in conjunction with the interaction server system **110** provides for a multi-clip capture mode. The multi-clip capture mode is optional and can be selected for activation by the user. The multi-clip capture mode corresponds with capturing multiple video clips or pictures which are combinable to generate a media content item (e.g., for sending to a friend, broadcasting to others, etc.). The multi-clip capture mode provides for a preview bar in which each of the multiple video clips is represented by a thumbnail, and the thumbnails are presented in a combined clip. In example embodiments, the combined clip includes front and back handles for trimming the entirety of the combined clip. Moreover, the combined clip is selectable by the user, to trim (e.g., via front and back handles for

the individual clip), reorder and/or delete individual video clips.

[0106] In the multi-clip capture mode, each clip is a separate video recording (e.g., associated with a respective press-and-hold gesture) or picture (e.g., associated with a respective tap gesture) taken by the user. Thus, with respect to FIG. 5, the interaction client **104** is configured to display the message preview interface (e.g., at block 524) upon user completion of the combined clips. For example, the interaction client **104** provides a message preview button (e.g., an “edit and send” button) for redirecting to the message preview interface.

[0107] At block 518, the interaction client **104** provides for the shutter button to be draggable in different directions to perform respective functions related to video recording. By way of non-limiting example, the different directions include: a first direction (e.g., left) to initiate hands-free recording; a second direction (e.g., right) to switch between a front-facing camera and a rear-facing camera of the device; a third direction (e.g., upward) to zoom in on the real-time image data; and a fourth direction (e.g., downward) to zoom out from the real-time image data.

[0108] With respect to the first draggable direction, the interaction client **104** provides for the shutter button to be draggable in this direction during video recording (e.g., while the press-and-hold gesture is still being held), to initiate hands-free recording. As described herein, hands-free recording provides for the interaction client **104** to continue video recording with out the user continually holding the shutter button **1114**. In example embodiments, the first direction is dragging the shutter button towards the hands-free icon (e.g., in the left direction).

[0109] In example aspects, the interaction client **104** is configured to detect when the drag gesture meets a preset condition, to prompt the user for hands-free recording. For example, the preset condition corresponds to the shutter button being positioned at least partially on top of the hands-free icon (e.g., based on threshold distances and/or overlap of predefined pixels between the shutter button and the hands-free icon). In this manner, the shutter button can be continually dragged, for example in multiple directions, with hands-free recording being prompted to the user each time the preset condition is met.

[0110] In a case where the preset condition is met (e.g., while the shutter button is held), the interaction client **104** is configured to prompt the user by displaying a visual hint (e.g., “release to record hands free”). The visual hint notifies the user that hands-free recording can be initiated by releasing the shutter button. The visual hint is toggled on/off as movement of the shutter button meets and does not meet the preset condition. Upon detecting release of the shutter button while the preset condition is met, the interaction client **104** initiates hands-free recording such that video recording continues while the user is no longer holding the shutter button.

[0111] In example embodiments, the interaction client **104** updates the shutter button upon activation of hands-free recording. For example, the interaction client **104** displays an animation which causes the shutter button to return back to its original position (e.g., from bottom-left to bottom-center). In addition, the interaction client **104** causes the shutter button to change from a circle of solid color (e.g., yellow) to a square of the same solid color. In example aspects, the square shape indicates that the shutter button is user-selectable to stop video recording. In example aspects, the interaction client **104** is configured to depict the respective icon for an augmented reality content item, if activated, as centered within the shutter button.

[0112] Regarding the second draggable direction, the interaction client **104** provides for the shutter button to be draggable in this direction during video recording (e.g., while the press-and-hold gesture is still being held), to switch between the front-facing camera and the rear-facing camera of the user system **102**. In example embodiments, the second direction is dragging the shutter button towards the flip camera icon (e.g., in the right direction).

[0113] In example aspects, the interaction client **104** is configured to detect when the drag gesture meets a preset condition so as to switch between the front-facing camera and the rear-facing camera (or vice versa). For example, the preset condition corresponds to the shutter button being positioned at least partially on top of the flip camera icon (e.g., based on threshold distances and/or

overlap of predefined pixels between the shutter button and the flip camera icon). In this manner, the shutter button can be continually dragged, for example in multiple directions, with the cameras switching between front-facing and rear-facing each time the preset condition is met.

[0114] Regarding the third and fourth draggable directions, the interaction client **104** provides for the shutter button to be draggable in these directions during video recording (e.g., while the press-and-hold gesture is still being held), to respectively zoom in and zoom out. In example embodiments, the third direction is dragging the shutter button upwards, and the fourth direction is dragging the shutter button downwards.

[0115] In example aspects, the shutter button can be continually dragged, for example in multiple directions, with the interaction client **104** performing a zoom in for each drag gesture in the third direction (e.g., upwards) and a zoom out for each drag gesture in the fourth direction (e.g., downwards).

[0116] At decision block **520**, the interaction client **104** determines whether the user has stopped video recording. In a case where the multi-clip capture mode is disabled and hands-free recording is not activated, the interaction client **104** determines that the user selected to stop video recording by detecting release of the press-and-hold gesture. In a case where the multi-clip capture mode is enabled, the interaction client **104** determines that the user selected to stop video recording by detecting user selection of the message preview button (e.g., the “edit and send” button). In a case where hands-free is activated, the interaction client **104** determines that the user selected to stop video recording by detecting user selection of the shutter button (e.g., which is square-shaped during hands-free recording and functions as a stop recording button).

[0117] If the user has not stopped video recording, the interaction client **104** continues to provide the indications that video recording is active per block **516** and to provide for the shutter button to be draggable in different directions to perform respective functions per block **518**. As shown in the example of FIG. 5, blocks **516-520** are repeated until the user stops video recording. Upon a determination that the user stops video recording at decision block **520**, the interaction client **104** causes the video to stop recording (block **522**). The interaction client **104** then displays a message preview interface (block **524**).

[0118] As noted, the message preview interface provides for a media content item to be generated based on the video recording. In addition, the message preview interface includes editing tools for modifying/annotating (e.g., drawing on, adding text to, adding stickers to, cropping, and the like) the video recording. Moreover, the message preview interface includes interface elements (e.g., buttons) for one or more of: saving the video recording (e.g., with modifications/annotations) as a media content item; creating or updating a Story based on the video recording (e.g., with modifications/annotations); modifying audio signal(s) associated with the video recording; sending a media content item which includes the video recording (e.g., with modifications/annotations) to a contact/friend; and/or broadcasting the media content item in association with a feed interface (e.g., for viewing by other users who are not necessarily contacts/friends).

[0119] For multi-clip video capture, the message preview interface includes elements for one or more of: modifying/annotating (e.g., drawing on, adding text to, adding stickers to, cropping, and the like) each of the captured video clips; combining the captured video clips (e.g., with modifications/annotations) and saving the combined video clips as a single media content item; creating or updating a Story based on the captured video clips (e.g., with modifications/annotations); modifying audio signal(s) associated with each of the captured video clips; sending a media content item which includes the captured video clips (e.g., with modifications/annotations) to a contact/friend; and/or broadcasting the media content item in association with a feed interface. The process then ends at end block **526**.

[0120] On the other hand, in response to determining that the user selection is not a press-and-hold gesture (e.g., and is instead a tap gesture) at decision block **508**, the interaction client **104** at block **510** causes the user system **102** to take a picture. As noted above, the picture is based on the image

data that is captured by the device screen and displayed in real-time on the device screen (e.g., together with augmented reality content, if applicable).

[0121] At block **512**, the interaction client **104** provides an indication of taking the picture within the user interface. For example, the interaction client **104** replaces the above-noted first set of interface elements (e.g., profile button, search button, memories button, carousel interface launch button, toolbar and/or tab bar) with a subset of the second set of interface elements. For example, the interaction client **104** still presents the above-described bounce animation, which scales down the camera viewport from its original size to a smaller size, scales up the camera viewport from the smaller size to its original size), and displays the border upon completion of the bounce animation. However, the subset of the second set of interface elements no longer includes one or more of the animation icon, the hands-free icon and/or the flip camera icon.

[0122] After taking the picture, the interaction client **104** displays the message preview interface (block **524**). With respect to taking a picture, the message preview interface includes elements for one or more of: modifying/annotating the picture; saving the picture (e.g., with modifications/annotations) as a media content item; creating or updating a Story based on the picture (e.g., with modifications/annotations); modifying audio signal(s) associated with the picture; sending a media content item which includes the picture (e.g., with modifications/annotations) to a contact/friend; and/or broadcasting the media content item in association with a feed interface. The process **500** then ends at end block **526**.

[0123] FIGS. **6A-6B** illustrate an example user interface **602** for providing an indication of video recording, in accordance with some examples. In the example of FIG. **6A**, the user interface **602** includes image data **604**, a profile button **606**, a search button **608**, a toolbar **610** (which includes a flip camera button **612**, a flash button **614**, a night mode button **616** and an additional tools button **618**), a memories button **620**, a shutter button **622**, a carousel interface launch button **624**, and a tab bar **626** (which includes a map tab **628**, a chat tab **630**, a camera tab **632**, a stories tab **634** and a discover tab **636**).

[0124] For example, the user interface **602** of FIG. **6A** corresponds to a camera interface which is presented upon startup of the interaction client **104**. The interaction client **104** activates a camera (e.g., a front-facing camera, or a rear-facing camera) upon startup. The user interface **602** displays the captured image data **604** captured by the camera in real-time.

[0125] As noted above with respect to FIG. **5**, the user interface **602** includes a first set of interface elements in conjunction with the real-time image data. For example, the profile button **606** is user-selectable to surface a user profile, and the search button **608** is user-selectable to perform various types of searches within the interaction client **104**.

[0126] In example embodiments, the first set of interface elements also includes the memories button **620** which is user-selectable to access saved media content items, and the carousel interface launch button **624** which is user-selectable to surface a carousel interface for augmented reality content. As noted above, the carousel interface (not shown) allows the user to cycle through and/or select different augmented reality content items, represented by respective icons, to apply with respect to the image data **604**. Each augmented reality content item provides for adding a real-time special effect and/or sound to the image data **604**. In example embodiments, the toolbar **610** includes the flip camera button **612** for switching between front and rear facing cameras, the flash button **614** for activating a flash of the camera, the night mode button **616** for enhancing dark scenes, and the additional tools button **618** for accessing additional modes/features (e.g., including the multi-clip capture mode).

[0127] Moreover, the tab bar **626** provides for navigating between tabs of the interaction client **104**. By way of non-limiting example, the tab bar **626** includes the map tab **628** which is user-selectable to switch to a map interface which displays user icons or avatars on a map to indicate a current or past location of friends of a user, as well as media content generated by such friends, within the context of a map. The chat tab **630** is user-selectable to switch to a chat interface for

communicating between friends of the user. The camera tab **632** is user-selectable to switch to the camera interface (e.g., corresponding to the user interface **602**). The stories tab **634** is user-selectable to switch to a stories interface for accessing content collections (e.g., stories) from friends of the user, subscriptions and recommendations. Moreover, the discover tab **636** is user-selectable to switch to a discover interface for accessing original video content (e.g., as provided by an administrator of the interaction system **100**), content collections (e.g., stories) that are currently trending on the interaction system **100**, and/or content collections (e.g., stories) covering politics, entertainment, sports and more as provided by predetermined publishers (e.g., corporate partners). [0128] As discussed above with respect to FIG. 5, the shutter button **622** is user-selectable to take a picture (e.g., in response to a press/tap gesture) or to record a video (e.g., in response to a press-and-hold gesture) of image data captured by the camera. In the example of FIGS. 6A-6B, the user provides touch input of a press-and-hold gesture. In response to this gesture, the interaction client **104** provides different indications that video recording is active as shown in FIG. 6B.

[0129] In the example of FIG. 6B, the user interface **602** provides indications that video recording is active. For example, the user interface **602** replaces the first set of interface elements (e.g., elements **606-620** and **624-636** of FIG. 6A) with a second set of interface elements (e.g., elements **638-644** of FIG. 6B). In example embodiments, the interaction client **104** causes the first set of interface elements to fade out in association with being replaced by the second set of interface elements.

[0130] As noted above, the border **638** is presented as a solid color (e.g., yellow) along the periphery of the camera viewport for the image data **604**. In example embodiments, the border is scaled up within the user interface **602**. While not shown in the example of FIG. 6B, the user interface **602** may include a ring flash effect (e.g., glowing halo) along the periphery of the camera viewport, with the border **638** being displayed on top of the ring flash effect. The user interface **602** presents a bounce animation with respect to the camera viewport, with the border **638** being displayed upon completion of the bounce animation.

[0131] In example embodiments, the user interface **602** scales up the animated icon **640** within a predefined area (e.g., top center and within the border **638**). The animated icon **640** may scale up to a predefined size (e.g., 30×30 pixels), and blink every predefined interval (e.g., every second). For example, the blinking includes changing opacity from 0% to 100% in 0.4 s, pausing for 3 s, and changing opacity from 100% to 0% in 0.3 s. In the example of FIG. 6B, the animated icon **640** is a ghost icon, but other icons may be used instead (e.g., a smiley face icon, a company logo, or another type of icon).

[0132] With respect to the hands-free icon **642**, the user interface **602** scales up this icon within a predefined area (e.g., to the left of the shutter button **622**). In the example of FIG. 6B, the hands-free icon **642** is depicted as a lock. The user may activate hands-free recording, with respect to the currently-active video recording, via a predefined gesture. For example, the predefined gesture is a slide left gesture while still holding the press-and-hold gesture.

[0133] Regarding the flip camera icon **644**, the user interface **602** scales up this icon within a predefined area (e.g., to the right of the shutter button **622**). The user may switch between the rear and front facing devices cameras, while still recording the current video, via a predefined gesture. For example, the predefined gesture is a slide right gesture while still holding the press-and-hold gesture.

[0134] As noted above, the user interface **602** also updates the shutter button **622** to indicate that video recording is active. In a case where no augmented reality content item is selected, the shutter button **622** initially appears as a circle with a border (e.g., white border) and no fill (e.g., shown in FIG. 6A). In the example of FIG. 6B, the user interface **602** updates the appearance of the shutter button **622** by scaling up the circle and changing the entire circle into a solid color (e.g., yellow).

[0135] FIG. 7 illustrates an example user interface **700** for providing an indication of video recording together with augmented reality content, in accordance with some examples. Similar to

the user interface **602** of FIG. **6B**, the user interface **700** of FIG. **7** includes image data **702**, a border **704**, an animated icon **706**, a hands-free icon **708**, a shutter button **710** and a flip camera icon **712**.

[0136] However, the example of FIG. **7** illustrates an example scenario in which the user selected an augmented reality content item via a carousel interface. As noted above, the carousel interface (not shown) is surfaced in response to user selection of the carousel interface launch button (e.g., element **624** in FIG. **6B**). The carousel interface present a row of icons, each representing a respective augmented reality content item for applying to the captured image data **702**. The icon corresponding to an active augmented reality content item (e.g., which applies a real-time effect/sound to the captured image data) is displayed as centered within the shutter button **710**.

[0137] In the example of FIG. **7**, the user selects to perform a press-and-hold gesture for video recording the captured image data **702**. Thus, the shutter button **710** which already includes the respective icon centered within the circle (e.g., having a border in a first color such as white) is scaled up (e.g., the circle grows larger). In addition, the border of the of circle changes color (e.g., to a second color such as yellow). The icon centered within the shutter button **710** is maintained during the press-and-hold gesture.

[0138] FIG. **8** illustrates an example user interface **800** for providing an indication of video recording together with a preview icon, in accordance with some examples. Similar to the user interface **602** of FIG. **6B**, the user interface **800** of FIG. **8** includes image data **802**, a border **804**, an animated icon **806**, a hands-free icon **808**, a shutter button **810** and a flip camera icon **812**.

[0139] As discussed further below with respect to FIG. **10**, updating the appearance of the shutter button **810** includes presenting a progress indicator along a periphery of the shutter button **810**. In example embodiments, the interaction client **104** provides for each full rotation of the carved-out segment around the circle to take a preset time (e.g., **10** seconds to travel around the circle).

[0140] In the example of FIG. **8**, upon completion of each full rotation, the user interface **800** adds a preview icon **814** associated with that video segment having the preset time length (e.g., **10** seconds). As noted above, the preview icon is a single video frame within the video segment that is automatically selected by the interaction client **104**. Additional video segments of the preset time length (e.g., **10** seconds) are added as preview icons within the user interface (not shown), to create a series of video segments. For example, the border **804** provides a visual approximation of the length of the video during recording (e.g., approximated as the number of preview icons×the preset time length).

[0141] FIG. **9** illustrates an example user interface **900** for providing an indication of video recording in association with a multi-clip capture mode, in accordance with some examples. Similar to the user interface **602** of FIG. **6B**, the user interface **900** of FIG. **9** includes image data **902**, a border **904**, an animated icon **906**, a hands-free icon **908**, a shutter button **910** and a flip camera icon **912**.

[0142] As noted above with respect to FIG. **5**, the interaction client **104** in conjunction with the interaction server system **110** provides for a multi-clip capture mode. The multi-clip capture mode is optional and can be selected for activation by the user (e.g., as an option accessible via the additional tools button **618** of FIG. **6A**). The multi-clip capture mode corresponds with capturing multiple video clips or pictures which are combinable to generate a media content item (e.g., for sending to a friend, broadcasting to others, etc.).

[0143] The multi-clip capture mode provides for a preview bar (not shown) in which each of the multiple video clips is represented by a thumbnail, and the thumbnails are presented in a combined clip. In example embodiments, the combined clip includes front and back handles for trimming the entirety of the combined clip. Moreover, the combined clip is selectable by the user, to trim (e.g., via front and back handles for the individual clip), reorder and/or delete individual video clips.

[0144] In the multi-clip capture mode, each clip is a separate video recording (e.g., associated with a respective press-and-hold gesture) or picture (e.g., associated with a respective tap gesture) taken

by the user. The interaction client **104** is configured to display the message preview interface upon user completion of the combined clips. For example, the user interface **900** includes a message preview button **914** (e.g., an “edit and send” button) for redirecting to the message preview interface.

[0145] With respect to the captured video clips, the message preview interface includes elements for one or more of: modifying/annotating the each of the captured video clips; combining the captured video clips and saving the combined video clips as a single media content item; creating or updating a Story based on the captured video clips; modifying audio signal(s) associated with the captured video clips; sending a media content item which includes the captured video clips to a contact/friend; and/or broadcasting the media content item in association with a feed interface.

[0146] FIG. **10** illustrates example user interfaces **1002-1008** for indicating progress of video recording, in accordance with some examples. For example, each of the user interfaces **1002**, **1004**, **1006** and **1008** corresponds to a different treatment (e.g., version) of showing video recording progress with respect to respective shutter buttons **1010**, **1012**, **1014** and **1016**. In example embodiment, the treatment (e.g., version) is set by an administrator of the interaction system **100**.

[0147] As noted above, updating the appearance of the shutter button includes presenting a progress indicator along a periphery of the shutter button. For example, the user interface **1004** depicts the shutter button **1012** with a circle in a first color (e.g., yellow) and a progress indicator in a second color (e.g., gray). The progress indicator is displayed as an animation in which a carved-out segment positioned at the edge of the circle, travels around the circle in a clockwise manner and increases in size. Each full rotation of the carved-out segment around the circle takes a preset time (e.g., 10 seconds to travel around the circle), with the carved-out segment resetting after each rotation.

[0148] In another example, the user interface **1006** depicts the shutter button **1014** with the circle and progress indicator having the same color (e.g., yellow). Instead of a carved-out segment, the progress indicator is depicted as a growing ring that travels around the circle and resets after each full rotation. In another example, the user interface **1008** depicts the shutter button **1016** with the circle in a first color (e.g., gray) and the progress indicator in second color (e.g., gray). The progress indicator is depicted as a growing ring that travels around the circle and resets after each full rotation. In yet another example, the user interface **1002** depicts the shutter button without a progress indicator.

[0149] FIGS. **11A-11B** illustrate an example user interface **1102** for providing a shutter button to be draggable in a first direction to initiate hands-free recording, in accordance with some examples. Similar to the user interface **602** of FIG. **6B**, the user interface **1102** includes image data **1104**, a border **1106**, an animated icon **1108**, a hands-free icon **1110**, a flip camera icon **1112** and a shutter button **1114**.

[0150] FIG. **11A** depicts touch input **1116** of the shutter button **1114**. For example, the touch input **1116** corresponds to a press-and-hold gesture that initiated video recording, resulting in the display of elements **1106-1112** and in the updated display of the shutter button **1114**. For illustrative purposes, the touch input is depicted as a smaller circle within the shutter button **1114**.

[0151] As noted above, the interaction client **104** provides for the shutter button **1114** to be draggable in a first direction during video recording (e.g., while the press-and-hold gesture is still being held), to initiate hands-free recording. As noted above, hands-free recording provides for the interaction client **104** to continue video recording without the user continually holding the shutter button **1114**. In example embodiments, the first direction is dragging the shutter button **1114** towards the hands-free icon **1110** (e.g., in the left direction).

[0152] Thus, FIG. **11A** illustrates an example scenario in which the user drags the shutter button **1114** in such first direction. For illustrative purposes, the drag gesture is depicted as a left arrow. The interaction client **104** is configured to depict the shutter button **1114** as moving along with the touch input **1116** during the drag gesture. In addition, the interaction client **104** is configured to

detect when the drag gesture meets a preset condition so as to prompt the user for hands-free recording.

[0153] For example, the preset condition corresponds to the shutter button **1114** being positioned at least partially on top of the hands-free icon **1110** (e.g., based on threshold distances and/or overlap of predefined pixels between the shutter button **1114** and the hands-free icon **1110**). In this manner, the shutter button **1114** can be continually dragged, for example in multiple directions, with hands-free recording being prompted to the user when the preset condition is met. In example embodiments, the hands-free icon **1110** changes from a first color (e.g., white) to a second color (e.g., black) when the preset condition is met. Moreover, the shutter button **1114** scales down (e.g., to 80%) while the preset condition is met, and scales back up to its original size when the preset condition is no longer met. In a case where the user releases the press-and-hold gesture while the preset condition is not met, the interaction client **104** stops the video recording and displays the message preview interface (e.g., per blocks **520-524** in FIG. 5).

[0154] The example of FIG. **11B** illustrates a case where the preset condition is met while the shutter button **1114** is held. In response, the interaction client **104** is configured to prompt the user, by displaying a visual hint (e.g., “release to record hands-free”). The visual hint notifies the user that hands-free recording can be initiated by releasing the shutter button **1114**. The visual hint is toggled on/off as movement of the shutter button **1114** meets and does not meet the preset condition. Upon detecting release of the shutter button **1114** while the preset condition is met, the interaction client **104** initiates hands-free recording such that video recording continues while the user is no longer holding the shutter button **1114**.

[0155] In example embodiments, the interaction client **104** updates the shutter button **1114** upon activation of hands-free recording. For example, the interaction client **104** displays an animation which causes the shutter button **1114** to return back to its original position (e.g., from bottom-left to bottom-center). In addition, the interaction client **104** causes the shutter button **1114** to change from a circle of solid color (e.g., yellow) to a square of the same solid color. In example aspects, the square shape indicates that the shutter button **1114** is user-selectable to stop video recording. Upon detecting a predefined gesture (e.g., a tap gesture) in association with the square-shaped shutter button **1114**, the interaction client **104** stops the video recording and displays the message preview interface (e.g., per blocks **520-524** in FIG. 5). While not shown in the example of FIGS. **11A-11B**, the interaction client **104** is configured to depict the respective icon for an augmented reality content item, if activated, as centered within the shutter button **1114** (e.g., in both the circular and square shape).

[0156] FIGS. **12A-12B** illustrate an example user interface **1202** for providing a shutter button to be draggable in a second direction to switch between a front-facing camera and a rear-facing camera, in accordance with some examples. Similar to the user interface **602** of FIG. **6B**, the user interface **1202** includes image data **1204**, a border **1206**, an animated icon **1208**, a hands-free icon **1210**, a flip camera icon **1212** and a shutter button **1214**.

[0157] FIG. **12A** depicts touch input **1216** of the shutter button **1214**. For example, the touch input **1216** corresponds to a press-and-hold gesture that initiated video recording, resulting in the display of elements **1206-1212** and in the updated display of the shutter button **1214**. For illustrative purposes, the touch input is depicted as a smaller circle within the shutter button **1214**.

[0158] As noted above, the interaction client **104** provides for the shutter button **1214** to be draggable in a second direction during video recording (e.g., while the press-and-hold gesture is still being held), to switch between the front-facing camera and the rear-facing camera of the user system **102**. In example embodiments, the second direction is dragging the shutter button **1214** towards the flip camera icon **1212** (e.g., in the right direction).

[0159] Thus, FIG. **12A** illustrates an example scenario in which the user drags the shutter button **1214** in such second direction. For illustrative purposes, the drag gesture is depicted as a right arrow. The interaction client **104** is configured to depict the shutter button **1214** as moving along

with the touch input **1216** during the drag gesture. In addition, the interaction client **104** is configured to detect when the drag gesture meets a preset condition so as to switch between the front-facing camera and the rear-facing camera (or vice versa).

[0160] With respect to FIG. **12B**, the preset condition corresponds to the shutter button **1214** being positioned at least partially on top of the flip camera icon **1212** (e.g., based on threshold distances and/or overlap of predefined pixels between the shutter button **1214** and the flip camera icon **1212**). In example embodiments, the flip camera icon **1212** changes from a first color (e.g., white) to a second color (e.g., black) when the preset condition is met. In this manner, the shutter button **1214** can be continually dragged, for example in multiple directions, with the cameras switching between front-facing and rear-facing each time the preset condition is met. In a case where the user selects to stop recording (e.g., by release of the press-and-hold gesture, or in association with hands-free recording or multi-clip capture), the interaction client **104** stops the video recording and displays the message preview interface (e.g., per blocks **520-524** in FIG. **5**).

[0161] FIGS. **13A-13B** illustrate an example user interface **1302** for providing a shutter button to be draggable in third and fourth directions to respectively zoom in and out, in accordance with some examples. Similar to the user interface **602** of FIG. **6B**, the user interface **1302** includes image data **1304**, a border **1306**, an animated icon **1308**, a hands-free icon **1310**, a flip camera icon **1312** and a shutter button **1314**.

[0162] FIG. **13A** depicts touch input **1316** of the shutter button **1314**. For example, the touch input **1316** corresponds to a press-and-hold gesture that initiated video recording, resulting in the display of elements **1306-1312** and in the updated display of the shutter button **1314**. For illustrative purposes, the touch input is depicted as a smaller circle within the shutter button **1314**.

[0163] As noted above, the interaction client **104** provides for the shutter button **1314** to be draggable in third and fourth directions during video recording (e.g., while the press-and-hold gesture is still being held), to respectively zoom in and zoom out. In example embodiments, the third direction is dragging the shutter button **1314** upwards, and the fourth direction is dragging the shutter button **1314** downwards. During a zoom in or zoom out operation, the shutter button **1314** is scaled down (e.g., to 80%) and returns to its original size upon being dragged to the original Y position. In example embodiments, the shutter button **1314** is not draggable down below the original Y position.

[0164] FIG. **13A** illustrates an example scenario in which the user drags the shutter button **1314** in the third direction (e.g., upward). For illustrative purposes, the drag gesture is depicted as an up arrow. The interaction client **104** is configured to depict the shutter button **1314** as moving along with the touch input **1316** during the drag gesture.

[0165] In the example of FIG. **13B**, the interaction client **104** provides for a zoom in operation with respect to the image data **1304**. While not shown in the example of FIG. **13B**, the user may subsequently select to drag the shutter button **1314** in the downward direction, thereby causing the interaction client **104** to perform a zoom out operation with respect to the image data **1304**. In this manner, the shutter button **1314** can be continually dragged, for example in multiple directions, with the interaction client **104** performing a zoom in for each drag gesture in the third direction and a zoom out for each drag gesture in the fourth direction. In a case where the user selects to stop recording (e.g., by release of the press-and-hold gesture, or in association with hands-free recording or multi-clip capture), the interaction client **104** stops the video recording and displays the message preview interface (e.g., per blocks **520-524** in FIG. **5**).

[0166] FIG. **14** illustrates example user interfaces **1402-1406** for customizing an accent color used to indicate that video recording is active, in accordance with some examples. For example, FIG. **14** depicts a personal profile interface **1402**, an exclusive features interface **1404** and a capturing color interface **1406**.

[0167] In example embodiments, the interaction system **100** provides selected users with exclusive features that are not available to all users of the interaction system **100**. For example, the selected

users are those users who opt into a subscription plan. The subscription plan corresponds to a paid plan (e.g., a monthly subscription) that provides access to the new and exclusive features. For example, the exclusive features allow for customizing different user interfaces, or interface elements, displayed by the interaction client **104**.

[0168] In example embodiments, the exclusive features include customization of the accent color used to indicate that video recording is active. For example, the above-described border, animated icon and shutter button are assigned a default color (e.g., yellow) with respect to video recording. The selected users are provided with user interfaces **1402-1406** for changing the default color, while other users of the interaction system **100** (e.g., those who did not opt into the subscription plan) are not provided with such user interfaces.

[0169] In the example of FIG. **14**, the interaction client **104** provides for a personal profile interface **1402**. For example, the personal profile interface **1402** is accessible via user selection of the profile button **606** discussed above with respect to FIG. **6A**. The personal profile interface **1402** presents information about the end user, such as user name, telephone number, address, and/or friends (e.g., contacts) of the end user. Moreover, the personal profile interface **1402** includes user-selectable options for updating account settings corresponding to the personal profile, such as personal avatar(s).

[0170] For selected users (e.g., those opting into the subscription plan), the personal profile interface **1402** provides additional options for updating account settings. As shown, the personal profile interface **1402** includes an exclusive features button **1408** for redirecting to the exclusive features interface **1404**.

[0171] The exclusive features interface **1404** includes several options/customizations available to the end user (e.g., who opted into the subscription plan). One such customization is accessible via the capturing color button **1410**. User selection of the capturing color button **1410** redirects to the capturing color interface **1406**.

[0172] As shown in the example of FIG. **14**, the capturing color interface **1406** allows the end user to select the accent color to be used when capturing video. As described herein, the accent color applies to the border, animated icon and shutter button used during video recording. In this manner, the interaction system **100** provides for the selected users to customize the accent color used during recording.

[0173] FIG. **15** is a flowchart illustrating a process **1500** for providing a draggable shutter button during video recording, in accordance with some examples. For explanatory purposes, the process **1500** is primarily described herein with reference to the interaction client **104** of FIG. **1**. However, one or more blocks (or operations) of the process **1500** may be performed by one or more other components, and/or by other suitable devices. Further for explanatory purposes, the blocks (or operations) of the process **1500** are described herein as occurring in serial, or linearly. However, multiple blocks (or operations) of the process **1500** may occur in parallel or concurrently. In addition, the blocks (or operations) of the process **1500** need not be performed in the order shown and/or one or more blocks (or operations) of the process **1500** need not be performed and/or can be replaced by other operations. The process **1500** may be terminated when its operations are completed. In addition, the process **1500** may correspond to a method, a procedure, an algorithm, etc.

[0174] The interaction client **104** displays a user interface within an application running on a device, the user interface presenting real-time image data captured by a camera of the device, the user interface including a shutter button which is configured to be selectable by a user to initiate video recording in response to a first user gesture (block **1502**).

[0175] The interaction client **104** initiates, upon detecting the first user gesture selecting the shutter button, video recording with respect to the real-time image data (block **1504**).

[0176] The interaction client **104** provides for the shutter button to be draggable in predefined directions to perform respective functions related to the video recording (block **1506**). the shutter

button is configured to be fixed in place prior to initiating the video recording.

[0177] The predefined directions include one or more of: a first direction (e.g., left) to initiate hands-free recording; a second direction (e.g., right) to switch between a front-facing camera and a rear-facing camera of the device; a third direction (e.g., upward) to zoom in on the real-time image data; and a fourth direction (e.g., downward) to zoom out from the real-time image data.

[0178] In example embodiments, the interaction client **104** provides, upon detecting the first user gesture selecting the shutter button, plural indications within the user interface that video recording is active.

[0179] In example embodiments, the first user gesture is a press-and-hold gesture, and the plural indications are provided for a duration of the press-and-hold gesture. Upon detecting a release of the press-and-hold gesture, the interaction client **104** causes the video recording to stop, and switches from the user interface to a second user interface for previewing and modifying the video recording.

[0180] In example embodiments, the shutter button is selectable to take a picture in response to a second user gesture. Upon detecting the second user gesture selecting the shutter button, the interaction client **104** takes the picture with respect to the real-time image data, and provides an indication of taking the picture within the user interface.

Machine Architecture

[0181] FIG. **16** is a diagrammatic representation of the machine **1600** within which instructions **1602** (e.g., software, a program, an application, an applet, an app, or other executable code) for causing the machine **1600** to perform any one or more of the methodologies discussed herein may be executed. For example, the instructions **1602** may cause the machine **1600** to execute any one or more of the methods described herein. The instructions **1602** transform the general, non-programmed machine **1600** into a particular machine **1600** programmed to carry out the described and illustrated functions in the manner described. The machine **1600** may operate as a standalone device or may be coupled (e.g., networked) to other machines. In a networked deployment, the machine **1600** may operate in the capacity of a server machine or a client machine in a server-client network environment, or as a peer machine in a peer-to-peer (or distributed) network environment. The machine **1600** may comprise, but not be limited to, a server computer, a client computer, a personal computer (PC), a tablet computer, a laptop computer, a netbook, a set-top box (STB), a personal digital assistant (PDA), an entertainment media system, a cellular telephone, a smartphone, a mobile device, a wearable device (e.g., a smartwatch), a smart home device (e.g., a smart appliance), other smart devices, a web appliance, a network router, a network switch, a network bridge, or any machine capable of executing the instructions **1602**, sequentially or otherwise, that specify actions to be taken by the machine **1600**. Further, while a single machine **1600** is illustrated, the term “machine” shall also be taken to include a collection of machines that individually or jointly execute the instructions **1602** to perform any one or more of the methodologies discussed herein. The machine **1600**, for example, may comprise the user system **102** or any one of multiple server devices forming part of the interaction server system **110**. In some examples, the machine **1600** may also comprise both client and server systems, with certain operations of a particular method or algorithm being performed on the server-side and with certain operations of the particular method or algorithm being performed on the client-side.

[0182] The machine **1600** may include processors **1604**, memory **1606**, and input/output I/O components **1608**, which may be configured to communicate with each other via a bus **1610**. In an example, the processors **1604** (e.g., a Central Processing Unit (CPU), a Reduced Instruction Set Computing (RISC) Processor, a Complex Instruction Set Computing (CISC) Processor, a Graphics Processing Unit (GPU), a Digital Signal Processor (DSP), an Application Specific Integrated Circuit (ASIC), a Radio-Frequency Integrated Circuit (RFIC), another processor, or any suitable combination thereof) may include, for example, a processor **1612** and a processor **1614** that execute the instructions **1602**. The term “processor” is intended to include multi-core processors that may

comprise two or more independent processors (sometimes referred to as “cores”) that may execute instructions contemporaneously. Although FIG. 16 shows multiple processors 1604, the machine 1600 may include a single processor with a single-core, a single processor with multiple cores (e.g., a multi-core processor), multiple processors with a single core, multiple processors with multiples cores, or any combination thereof.

[0183] The memory 1606 includes a main memory 1616, a static memory 1618, and a storage unit 1620, both accessible to the processors 1604 via the bus 1610. The main memory 1606, the static memory 1618, and storage unit 1620 store the instructions 1602 embodying any one or more of the methodologies or functions described herein. The instructions 1602 may also reside, completely or partially, within the main memory 1616, within the static memory 1618, within machine-readable medium 1622 within the storage unit 1620, within at least one of the processors 1604 (e.g., within the processor's cache memory), or any suitable combination thereof, during execution thereof by the machine 1600.

[0184] The I/O components 1608 may include a wide variety of components to receive input, provide output, produce output, transmit information, exchange information, capture measurements, and so on. The specific I/O components 1608 that are included in a particular machine will depend on the type of machine. For example, portable machines such as mobile phones may include a touch input device or other such input mechanisms, while a headless server machine will likely not include such a touch input device. It will be appreciated that the I/O components 1608 may include many other components that are not shown in FIG. 16. In various examples, the I/O components 1608 may include user output components 1624 and user input components 1626. The user output components 1624 may include visual components (e.g., a display such as a plasma display panel (PDP), a light-emitting diode (LED) display, a liquid crystal display (LCD), a projector, or a cathode ray tube (CRT)), acoustic components (e.g., speakers), haptic components (e.g., a vibratory motor, resistance mechanisms), other signal generators, and so forth. The user input components 1626 may include alphanumeric input components (e.g., a keyboard, a touch screen configured to receive alphanumeric input, a photo-optical keyboard, or other alphanumeric input components), point-based input components (e.g., a mouse, a touchpad, a trackball, a joystick, a motion sensor, or another pointing instrument), tactile input components (e.g., a physical button, a touch screen that provides location and force of touches or touch gestures, or other tactile input components), audio input components (e.g., a microphone), and the like.

[0185] In further examples, the I/O components 1608 may include biometric components 1628, motion components 1630, environmental components 1632, or position components 1634, among a wide array of other components. For example, the biometric components 1628 include components to detect expressions (e.g., hand expressions, facial expressions, vocal expressions, body gestures, or eye-tracking), measure biosignals (e.g., blood pressure, heart rate, body temperature, perspiration, or brain waves), identify a person (e.g., voice identification, retinal identification, facial identification, fingerprint identification, or electroencephalogram-based identification), and the like. The motion components 1630 include acceleration sensor components (e.g., accelerometer), gravitation sensor components, rotation sensor components (e.g., gyroscope).

[0186] The environmental components 1632 include, for example, one or cameras (with still image/photograph and video capabilities), illumination sensor components (e.g., photometer), temperature sensor components (e.g., one or more thermometers that detect ambient temperature), humidity sensor components, pressure sensor components (e.g., barometer), acoustic sensor components (e.g., one or more microphones that detect background noise), proximity sensor components (e.g., infrared sensors that detect nearby objects), gas sensors (e.g., gas detection sensors to detection concentrations of hazardous gases for safety or to measure pollutants in the atmosphere), or other components that may provide indications, measurements, or signals corresponding to a surrounding physical environment.

[0187] With respect to cameras, the user system **102** may have a camera system comprising, for example, front cameras on a front surface of the user system **102** and rear cameras on a rear surface of the user system **102**. The front cameras may, for example, be used to capture still images and video of a user of the user system **102** (e.g., “selfies”), which may then be augmented with augmentation data (e.g., filters) described above. The rear cameras may, for example, be used to capture still images and videos in a more traditional camera mode, with these images similarly being augmented with augmentation data. In addition to front and rear cameras, the user system **102** may also include a 360° camera for capturing 360° photographs and videos.

[0188] Further, the camera system of the user system **102** may include dual rear cameras (e.g., a primary camera as well as a depth-sensing camera), or even triple, quad or penta rear camera configurations on the front and rear sides of the user system **102**. These multiple cameras systems may include a wide camera, an ultra-wide camera, a telephoto camera, a macro camera, and a depth sensor, for example.

[0189] The position components **1634** include location sensor components (e.g., a GPS receiver component), altitude sensor components (e.g., altimeters or barometers that detect air pressure from which altitude may be derived), orientation sensor components (e.g., magnetometers), and the like.

[0190] Communication may be implemented using a wide variety of technologies. The I/O components **1608** further include communication components **1636** operable to couple the machine **1600** to a network **1638** or devices **1640** via respective coupling or connections. For example, the communication components **1636** may include a network interface component or another suitable device to interface with the network **1638**. In further examples, the communication components **1636** may include wired communication components, wireless communication components, cellular communication components, Near Field Communication (NFC) components, Bluetooth® components (e.g., Bluetooth® Low Energy), Wi-Fi® components, and other communication components to provide communication via other modalities. The devices **1640** may be another machine or any of a wide variety of peripheral devices (e.g., a peripheral device coupled via a USB).

[0191] Moreover, the communication components **1636** may detect identifiers or include components operable to detect identifiers. For example, the communication components **1636** may include Radio Frequency Identification (RFID) tag reader components, NFC smart tag detection components, optical reader components (e.g., an optical sensor to detect one-dimensional bar codes such as Universal Product Code (UPC) bar code, multi-dimensional bar codes such as Quick Response (QR) code, Aztec code, Data Matrix, Dataglyph, MaxiCode, PDF417, Ultra Code, UCC RSS-2D bar code, and other optical codes), or acoustic detection components (e.g., microphones to identify tagged audio signals). In addition, a variety of information may be derived via the communication components **1636**, such as location via Internet Protocol (IP) geolocation, location via Wi-Fi® signal triangulation, location via detecting an NFC beacon signal that may indicate a particular location, and so forth.

[0192] The various memories (e.g., main memory **1616**, static memory **1618**, and memory of the processors **1604**) and storage unit **1620** may store one or more sets of instructions and data structures (e.g., software) embodying or used by any one or more of the methodologies or functions described herein. These instructions (e.g., the instructions **1602**), when executed by processors **1604**, cause various operations to implement the disclosed examples.

[0193] The instructions **1602** may be transmitted or received over the network **1638**, using a transmission medium, via a network interface device (e.g., a network interface component included in the communication components **1636**) and using any one of several well-known transfer protocols (e.g., hypertext transfer protocol (HTTP)). Similarly, the instructions **1602** may be transmitted or received using a transmission medium via a coupling (e.g., a peer-to-peer coupling) to the devices **1640**.

Software Architecture

[0194] FIG. 17 is a block diagram 1700 illustrating a software architecture 1702, which can be installed on any one or more of the devices described herein. The software architecture 1702 is supported by hardware such as a machine 1704 that includes processors 1706, memory 1708, and I/O components 1710. In this example, the software architecture 1702 can be conceptualized as a stack of layers, where each layer provides a particular functionality. The software architecture 1702 includes layers such as an operating system 1712, libraries 1714, frameworks 1716, and applications 1718. Operationally, the applications 1718 invoke API calls 1720 through the software stack and receive messages 1722 in response to the API calls 1720.

[0195] The operating system 1712 manages hardware resources and provides common services. The operating system 1712 includes, for example, a kernel 1724, services 1726, and drivers 1728. The kernel 1724 acts as an abstraction layer between the hardware and the other software layers. For example, the kernel 1724 provides memory management, processor management (e.g., scheduling), component management, networking, and security settings, among other functionalities. The services 1726 can provide other common services for the other software layers. The drivers 1728 are responsible for controlling or interfacing with the underlying hardware. For instance, the drivers 1728 can include display drivers, camera drivers, BLUETOOTH® or BLUETOOTH® Low Energy drivers, flash memory drivers, serial communication drivers (e.g., USB drivers), WI-FI® drivers, audio drivers, power management drivers, and so forth.

[0196] The libraries 1714 provide a common low-level infrastructure used by the applications 1718. The libraries 1714 can include system libraries 1730 (e.g., C standard library) that provide functions such as memory allocation functions, string manipulation functions, mathematic functions, and the like. In addition, the libraries 1714 can include API libraries 1732 such as media libraries (e.g., libraries to support presentation and manipulation of various media formats such as Moving Picture Experts Group-4 (MPEG4), Advanced Video Coding (H.264 or AVC), Moving Picture Experts Group Layer-3 (MP3), Advanced Audio Coding (AAC), Adaptive Multi-Rate (AMR) audio codec, Joint Photographic Experts Group (JPEG or JPG), or Portable Network Graphics (PNG)), graphics libraries (e.g., an OpenGL framework used to render in two dimensions (2D) and three dimensions (3D) in a graphic content on a display), database libraries (e.g., SQLite to provide various relational database functions), web libraries (e.g., WebKit to provide web browsing functionality), and the like. The libraries 1714 can also include a wide variety of other libraries 1734 to provide many other APIs to the applications 1718.

[0197] The frameworks 1716 provide a common high-level infrastructure that is used by the applications 1718. For example, the frameworks 1716 provide various graphical user interface (GUI) functions, high-level resource management, and high-level location services. The frameworks 1716 can provide a broad spectrum of other APIs that can be used by the applications 1718, some of which may be specific to a particular operating system or platform.

[0198] In an example, the applications 1718 may include a home application 1736, a contacts application 1738, a browser application 1740, a book reader application 1742, a location application 1744, a media application 1746, a messaging application 1748, a game application 1750, and a broad assortment of other applications such as a third-party application 1752. The applications 1718 are programs that execute functions defined in the programs. Various programming languages can be employed to create one or more of the applications 1718, structured in a variety of manners, such as object-oriented programming languages (e.g., Objective-C, Java, or C++) or procedural programming languages (e.g., C or assembly language). In a specific example, the third-party application 1752 (e.g., an application developed using the ANDROID™ or IOS™ software development kit (SDK) by an entity other than the vendor of the particular platform) may be mobile software running on a mobile operating system such as IOS™, ANDROID™, WINDOWS® Phone, or another mobile operating system. In this example, the third-party application 1752 can invoke the API calls 1720 provided by the operating system 1712 to facilitate functionalities described herein.

Conclusion

[0199] Thus, the interaction system **100** as described herein provides for a shutter button to be draggable in different directions to perform respective functions associated with video recording. In response to a press-and-hold gesture of the shutter button, the interaction client **104** initiates video recording. The interaction client **104** also presents a hands-free icon (e.g., a lock icon) for hands-free video recording, and a flip camera icon for switching between front and rear-facing cameras while recording. The interaction client **104** provides for the shutter button to be draggable in: a first direction (e.g., left, towards the hands-free icon) to initiate hands-free recording; a second direction (e.g., right, towards the flip camera icon) to switch between the front-facing camera and the rear-facing camera; a third direction (e.g., upward) to zoom in on image data captured in real-time; and a fourth direction (e.g., downward) to zoom out with respect to the captured image data.

[0200] By virtue of providing a shutter button that is draggable in different directions for different functions, the interaction client **104** provides increased interactive feedback and facilitates video recording. Without providing such a draggable shutter button, performing the respective functions may be difficult and/or cumbersome during video recording. Thus, the interaction client **104** facilitates the capturing of videos, thereby saving time for the user (e.g., by reducing the number of retakes), and reducing computational resources/processing power for the interaction system **100**.

GLOSSARY

[0201] “Carrier signal” refers, for example, to any intangible medium that is capable of storing, encoding, or carrying instructions for execution by the machine and includes digital or analog communications signals or other intangible media to facilitate communication of such instructions. Instructions may be transmitted or received over a network using a transmission medium via a network interface device.

[0202] “Client device” refers, for example, to any machine that interfaces to a

[0203] communications network to obtain resources from one or more server systems or other client devices. A client device may be, but is not limited to, a mobile phone, desktop computer, laptop, portable digital assistants (PDAs), smartphones, tablets, ultrabooks, netbooks, laptops, multi-processor systems, microprocessor-based or programmable consumer electronics, game consoles, set-top boxes, or any other communication device that a user may use to access a network.

[0204] “Communication network” refers, for example, to one or more portions of a network that may be an ad hoc network, an intranet, an extranet, a virtual private network (VPN), a local area network (LAN), a wireless LAN (WLAN), a wide area network (WAN), a wireless WAN (WWAN), a metropolitan area network (MAN), the Internet, a portion of the Internet, a portion of the Public Switched Telephone Network (PSTN), a plain old telephone service (POTS) network, a cellular telephone network, a wireless network, a Wi-Fi® network, another type of network, or a combination of two or more such networks. For example, a network or a portion of a network may include a wireless or cellular network, and the coupling may be a Code Division Multiple Access (CDMA) connection, a Global System for Mobile communications (GSM) connection, or other types of cellular or wireless coupling. In this example, the coupling may implement any of a variety of types of data transfer technology, such as Single Carrier Radio Transmission Technology (1xRTT), Evolution-Data Optimized (EVDO) technology, General Packet Radio Service (GPRS) technology, Enhanced Data rates for GSM Evolution (EDGE) technology, third Generation Partnership Project (3GPP) including 3G, fourth-generation wireless (4G) networks, Universal Mobile Telecommunications System (UMTS), High Speed Packet Access (HSPA), Worldwide Interoperability for Microwave Access (WiMAX), Long Term Evolution (LTE) standard, others defined by various standard-setting organizations, other long-range protocols, or other data transfer technology.

[0205] “Component” refers, for example, to a device, physical entity, or logic having boundaries defined by function or subroutine calls, branch points, APIs, or other technologies that provide for

the partitioning or modularization of particular processing or control functions. Components may be combined via their interfaces with other components to carry out a machine process. A component may be a packaged functional hardware unit designed for use with other components and a part of a program that usually performs a particular function of related functions. Components may constitute either software components (e.g., code embodied on a machine-readable medium) or hardware components. A “hardware component” is a tangible unit capable of performing certain operations and may be configured or arranged in a certain physical manner. In various examples, one or more computer systems (e.g., a standalone computer system, a client computer system, or a server computer system) or one or more hardware components of a computer system (e.g., a processor or a group of processors) may be configured by software (e.g., an application or application portion) as a hardware component that operates to perform certain operations as described herein. A hardware component may also be implemented mechanically, electronically, or any suitable combination thereof. For example, a hardware component may include dedicated circuitry or logic that is permanently configured to perform certain operations. A hardware component may be a special-purpose processor, such as a field-programmable gate array (FPGA) or an application-specific integrated circuit (ASIC). A hardware component may also include programmable logic or circuitry that is temporarily configured by software to perform certain operations. For example, a hardware component may include software executed by a general-purpose processor or other programmable processors. Once configured by such software, hardware components become specific machines (or specific components of a machine) uniquely tailored to perform the configured functions and are no longer general-purpose processors. It will be appreciated that the decision to implement a hardware component mechanically, in dedicated and permanently configured circuitry, or in temporarily configured circuitry (e.g., configured by software), may be driven by cost and time considerations. Accordingly, the phrase “hardware component” (or “hardware-implemented component”) should be understood to encompass a tangible entity, be that an entity that is physically constructed, permanently configured (e.g., hardwired), or temporarily configured (e.g., programmed) to operate in a certain manner or to perform certain operations described herein. Considering examples in which hardware components are temporarily configured (e.g., programmed), each of the hardware components need not be configured or instantiated at any one instance in time. For example, where a hardware component comprises a general-purpose processor configured by software to become a special-purpose processor, the general-purpose processor may be configured as respectively different special-purpose processors (e.g., comprising different hardware components) at different times. Software accordingly configures a particular processor or processors, for example, to constitute a particular hardware component at one instance of time and to constitute a different hardware component at a different instance of time. Hardware components can provide information to, and receive information from, other hardware components. Accordingly, the described hardware components may be regarded as being communicatively coupled. Where multiple hardware components exist contemporaneously, communications may be achieved through signal transmission (e.g., over appropriate circuits and buses) between or among two or more of the hardware components. In examples in which multiple hardware components are configured or instantiated at different times, communications between such hardware components may be achieved, for example, through the storage and retrieval of information in memory structures to which the multiple hardware components have access. For example, one hardware component may perform an operation and store the output of that operation in a memory device to which it is communicatively coupled. A further hardware component may then, at a later time, access the memory device to retrieve and process the stored output. Hardware components may also initiate communications with input or output devices, and can operate on a resource (e.g., a collection of information). The various operations of example methods described herein may be performed, at least partially, by one or more processors that are temporarily configured (e.g., by software) or permanently configured to

perform the relevant operations. Whether temporarily or permanently configured, such processors may constitute processor-implemented components that operate to perform one or more operations or functions described herein. As used herein, “processor-implemented component” refers to a hardware component implemented using one or more processors. Similarly, the methods described herein may be at least partially processor-implemented, with a particular processor or processors being an example of hardware. For example, at least some of the operations of a method may be performed by one or more processors or processor-implemented components. Moreover, the one or more processors may also operate to support performance of the relevant operations in a “cloud computing” environment or as a “software as a service” (SaaS). For example, at least some of the operations may be performed by a group of computers (as examples of machines including processors), with these operations being accessible via a network (e.g., the Internet) and via one or more appropriate interfaces (e.g., an API). The performance of certain of the operations may be distributed among the processors, not only residing within a single machine, but deployed across a number of machines. In some examples, the processors or processor-implemented components may be located in a single geographic location (e.g., within a home environment, an office environment, or a server farm). In other examples, the processors or processor-implemented components may be distributed across a number of geographic locations.

[0206] “Computer-readable storage medium” refers, for example, to both machine-storage media and transmission media. Thus, the terms include both storage devices/media and carrier waves/modulated data signals. The terms “machine-readable medium,” “computer-readable medium” and “device-readable medium” mean the same thing and may be used interchangeably in this disclosure.

[0207] “Machine storage medium” refers, for example, to a single or multiple storage devices and media (e.g., a centralized or distributed database, and associated caches and servers) that store executable instructions, routines and data. The term shall accordingly be taken to include, but not be limited to, solid-state memories, and optical and magnetic media, including memory internal or external to processors. Specific examples of machine-storage media, computer-storage media and device-storage media include non-volatile memory, including by way of example semiconductor memory devices, e.g., erasable programmable read-only memory (EPROM), electrically erasable programmable read-only memory (EEPROM), FPGA, and flash memory devices; magnetic disks such as internal hard disks and removable disks; magneto-optical disks; and CD-ROM and DVD-ROM disks. The terms “machine-storage medium,” “device-storage medium,” “computer-storage medium” mean the same thing and may be used interchangeably in this disclosure. The terms “machine-storage media,” “computer-storage media,” and “device-storage media” specifically exclude carrier waves, modulated data signals, and other such media, at least some of which are covered under the term “signal medium.”

[0208] “Non-transitory computer-readable storage medium” refers, for example, to a tangible medium that is capable of storing, encoding, or carrying the instructions for execution by a machine.

[0209] “Signal medium” refers, for example, to any intangible medium that is capable of storing, encoding, or carrying the instructions for execution by a machine and includes digital or analog communications signals or other intangible media to facilitate communication of software or data. The term “signal medium” shall be taken to include any form of a modulated data signal, carrier wave, and so forth. The term “modulated data signal” means a signal that has one or more of its characteristics set or changed in such a manner as to encode information in the signal. The terms “transmission medium” and “signal medium” mean the same thing and may be used interchangeably in this disclosure.

[0210] “User device” refers, for example, to a device accessed, controlled or owned by a user and with which the user interacts to perform an action, or an interaction with other users or computer systems.

Claims

1. A system comprising: at least one processor; and a memory storing instructions that, when executed by the at least one processor, configure the at least one processor to perform operations comprising: causing display of a user interface within an application, the user interface presenting real-time image data captured by a device camera, the user interface including a shutter button which is configured to be selectable by a user to initiate video recording in response to a first user gesture; and upon detecting the first user gesture selecting the shutter button, initiating video recording with respect to the real-time image data, and providing for the shutter button to be draggable in a first predefined direction to initiate hands-free recording.
2. The system of claim 1, the operations further comprising: providing for the shutter button to be draggable in a second predefined direction to switch between a front-facing camera and a rear-facing camera.
3. The system of claim 1, the operations further comprising: providing for the shutter button to be draggable in a third direction to zoom in on the real-time image data and a fourth direction to zoom out from the real-time image data.
4. The system of claim 1, the operations further comprising: providing, upon detecting the first user gesture selecting the shutter button, plural indications within the user interface that video recording is active.
5. The system of claim 4, wherein the first user gesture comprises a press-and-hold gesture, and wherein the plural indications are provided for a duration of the press-and-hold gesture.
6. The system of claim 1, wherein the shutter button is configured to be fixed in place prior to initiating the video recording.
7. The system of claim 1, wherein the shutter button is selectable to take a picture in response to a second user gesture.
8. The system of claim 7, the operations further comprising, upon detecting the second user gesture selecting the shutter button: taking the picture with respect to the real-time image data; and providing an indication of taking the picture within the user interface.
9. A method, comprising: causing display of a user interface within an application, the user interface presenting real-time image data captured by a device camera, the user interface including a shutter button which is configured to be selectable by a user to initiate video recording in response to a first user gesture; and upon detecting the first user gesture selecting the shutter button, initiating video recording with respect to the real-time image data, and providing for the shutter button to be draggable in a first predefined direction to initiate hands-free recording.
10. The method of claim 9, further comprising: providing for the shutter button to be draggable in a second predefined direction to switch between a front-facing camera and a rear-facing camera.
11. The method of claim 9, further comprising: providing for the shutter button to be draggable in a third direction to zoom in on the real-time image data and a fourth direction to zoom out from the real-time image data.
12. The method of claim 9, further comprising: providing, upon detecting the first user gesture selecting the shutter button, plural indications within the user interface that video recording is active.
13. The method of claim 12, wherein the first user gesture comprises a press-and-hold gesture, and wherein the plural indications are provided for a duration of the press-and-hold gesture.
14. The method of claim 9, wherein the shutter button is configured to be fixed in place prior to initiating the video recording.
15. The method of claim 9, wherein the shutter button is selectable to take a picture in response to a second user gesture.
16. The method of claim 15, further comprising, upon detecting the second user gesture selecting

the shutter button: taking the picture with respect to the real-time image data; and providing an indication of taking the picture within the user interface.

17. A non-transitory computer-readable storage medium, the computer-readable storage medium including instructions that when executed by a computer, cause the computer to perform operations comprising: causing display of a user interface within an application, the user interface presenting real-time image data captured by a device camera, the user interface including a shutter button which is configured to be selectable by a user to initiate video recording in response to a first user gesture; and upon detecting the first user gesture selecting the shutter button, initiating video recording with respect to the real-time image data, and providing for the shutter button to be draggable in a first predefined direction to initiate hands-free recording.

18. The non-transitory computer-readable storage medium of claim 17, further comprising: providing for the shutter button to be draggable in a second predefined direction to switch between a front-facing camera and a rear-facing camera.

19. The non-transitory computer-readable storage medium of claim 17, further comprising: providing for the shutter button to be draggable in a third direction to zoom in on the real-time image data and a fourth direction to zoom out from the real-time image data.

20. The non-transitory computer-readable storage medium of claim 17, the operations further comprising: providing, upon detecting the first user gesture selecting the shutter button, plural indications within the user interface that video recording is active.
